

EXPERIMENT #1: Investigate DC characteristics of a MOSFET – NMOS & PMOS

NMOS:

Goal of this experiment is to obtain the current as a function of gate to source voltage in the linear region, i.e., at a small V_{DS} bias.

Connect NMOS transistor as shown in Figure 1. Set V_{DD} voltage at a low positive value, i.e., $V_{DD} \leq 0.1$ V. Sweep V_{GG} voltage in the range from 0 V to +3 V. Have a small increment of about 0.1 V in the 0 to 1.5 V range and a larger one, e.g., 0.25 V in the rest of the range. Measure the drain to source current I_{DS} . Complete the table shown in next slide.

Observe at what voltage the drain to source current starts flowing.

Record the gate current at $V_{GG} = 1$ V and 3V

$V_{GG} = 1$ V $I_G =$ _____

$V_{GG} = 3$ V $I_G =$ _____

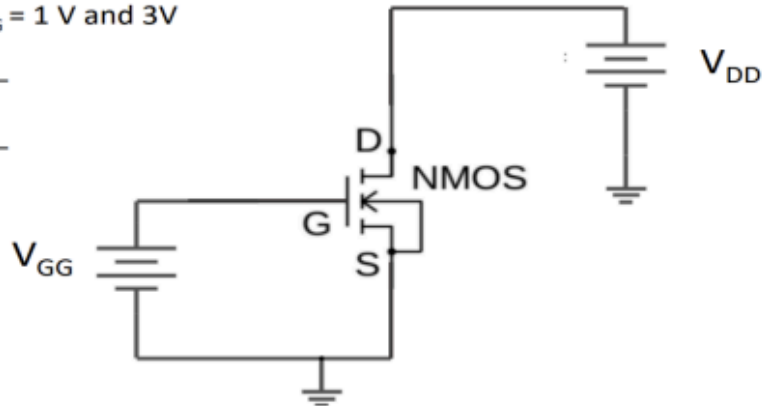


Figure 1

Set $V_{DD} = 0.1$ V

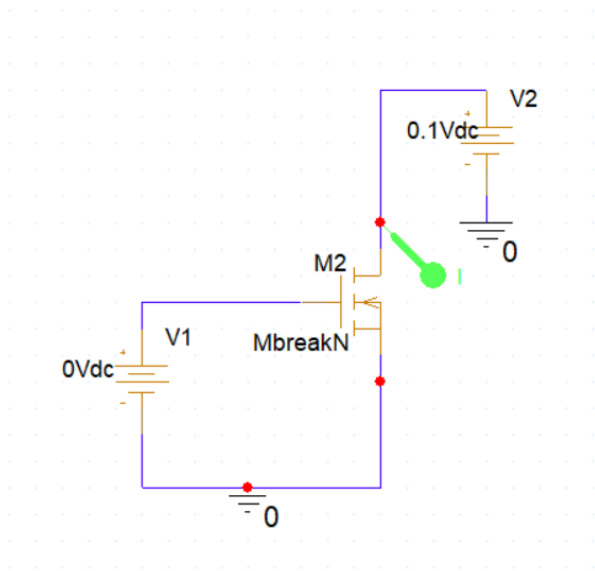
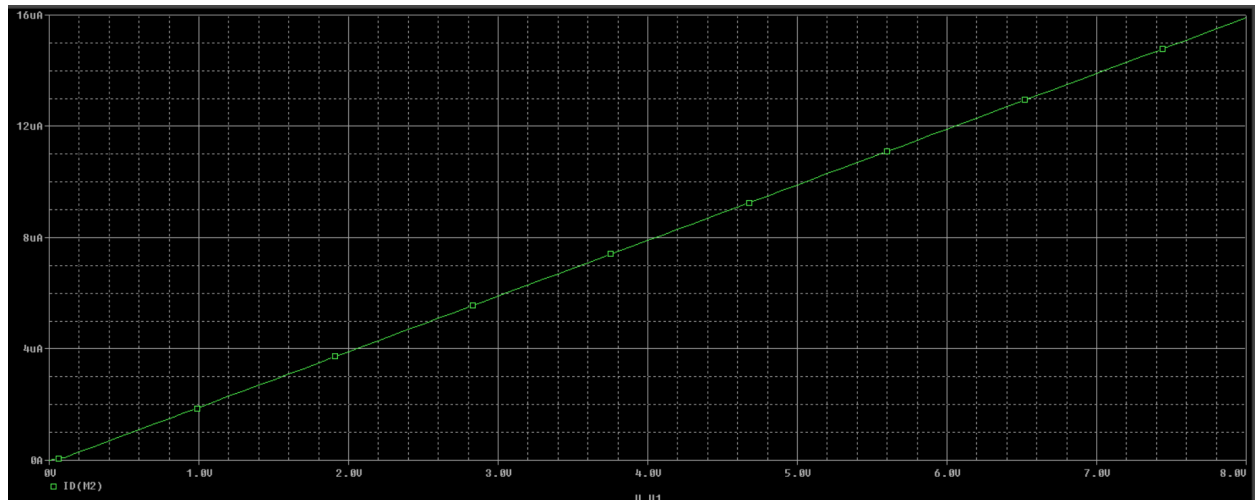
$V_{DD} =$ _____

NMOS Experiment 1 Measured Results

Experiment 1		
NMOS=2n7000FS		Mesured
	VGG=1V	IG=0.0006mA
	VGG=3V	IG=0.0005mA
	VDD=0.1V	VDD=0.10711V

VGG (V)	Actual VGG	IDS
0	0.266mV	0A
0.1	0.10781V	0A
0.2	0.18959V	0A
0.3	0.2989V	0A
0.4	0.3514V	0A
0.5	0.46892V	0A
0.6	0.55702V	0A
0.7	0.67971V	0A
0.8	0.78541V	0A
0.9	0.90577V	0A
1	1.00875V	0A
1.1	1.09322V	0A
1.2	1.1950V	0A
1.3	1.311V	0A
1.4	1.3659V	0A
1.5	1.4729V	0A
1.75	1.7526V	0.008mA
2	2.0058V	0.211mA
2.25	2.2671V	2.155mA
2.5	2.4956V	7.538mA
2.75	2.7732V	12.000mA
3	2.9932V	14.274mA

NMOS Experiment 1 Simulated Results



PMOS:

Goal of this experiment is to obtain the current as a function of gate to source voltage in the linear region, i.e., at a small V_{DS} bias.

Connect a PMOS transistor as shown in Figure 2. Set V_{DD} voltage at a low negative value, i.e., $V_{DD} \leq -0.1$ V. Sweep V_{GG} voltage in the range from 0 V to -3 V. Have a small increment of -0.1 V in the 0 to -1.5 V range and a larger one, e.g., -0.25 V in the rest of the range. Measure the drain to source current I_{DS} . Complete the table shown in next slide.

Observe at what voltage the drain to source current starts flowing.

Record the gate current at $V_{GG} = -1$ V and -3 V

$V_{GG} = -1$ V $I_G =$ _____

$V_{GG} = -3$ V $I_G =$ _____

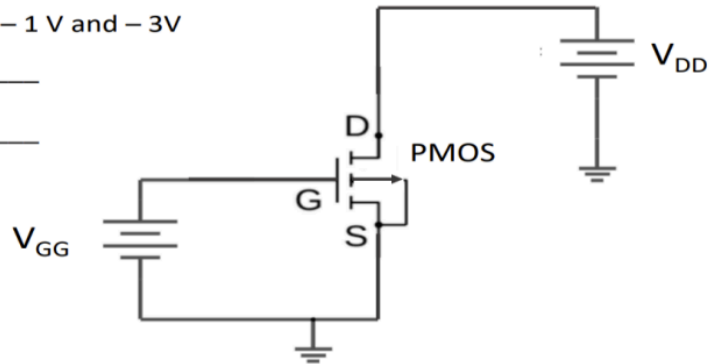
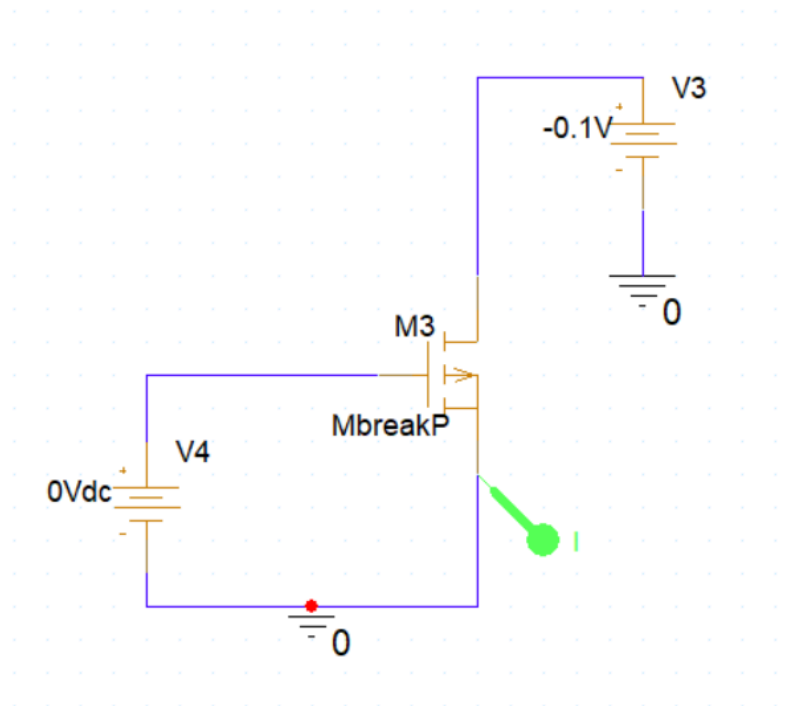
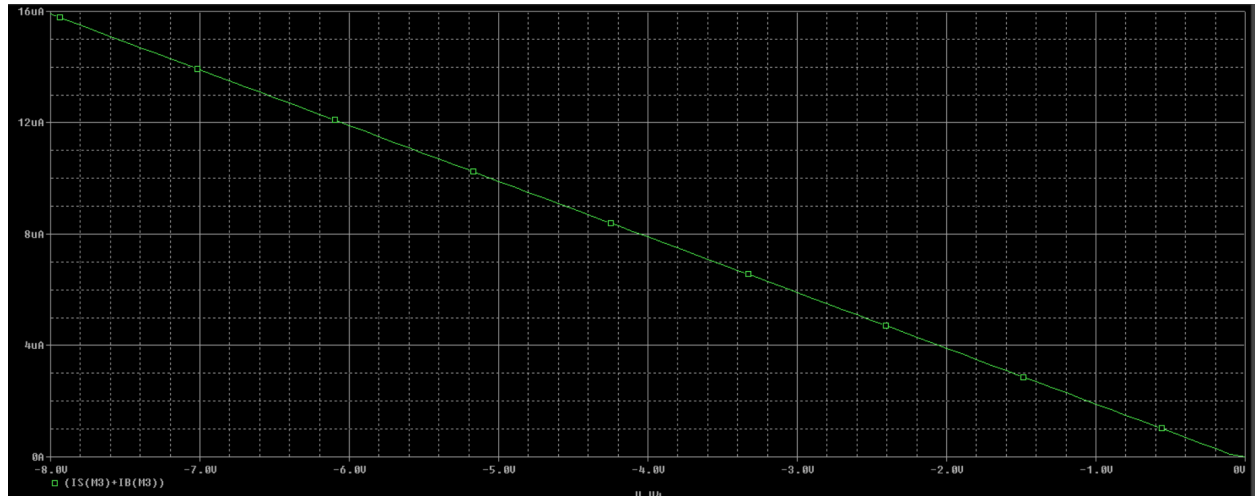


Figure 2

Experiment 1: PMOS simulation results





Conclusion: This experiment was a success due to the close correlation between the nmos measured and simulated values matching the graphs. The PMOS graph matched the graph that Professor Naher put in the slideshow for this experiment.

EXPERIMENT #2: Determine NMOS & PMOS output characteristics with a curve tracer.

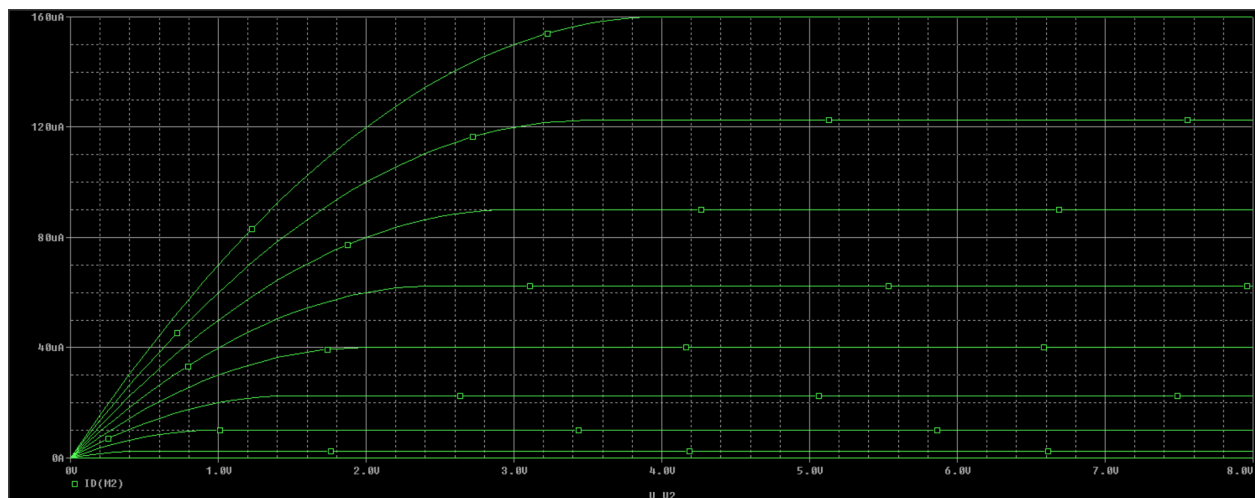
NMOS:

Goal of this experiment is to obtain the NMOS transistor output characteristics, i.e., drain to source current (I_{DS}) as a function of drain to source voltage (V_{DS}) for various gate to source voltages (V_{GS}).

Sweep V_{DS} voltage in the range from 0 V to +8 V, at an increment of 0.1 V.

Sweep V_{GS} voltage in the range from 0 V to +4 V at an increment of +0.5 V.

Record the characteristics.



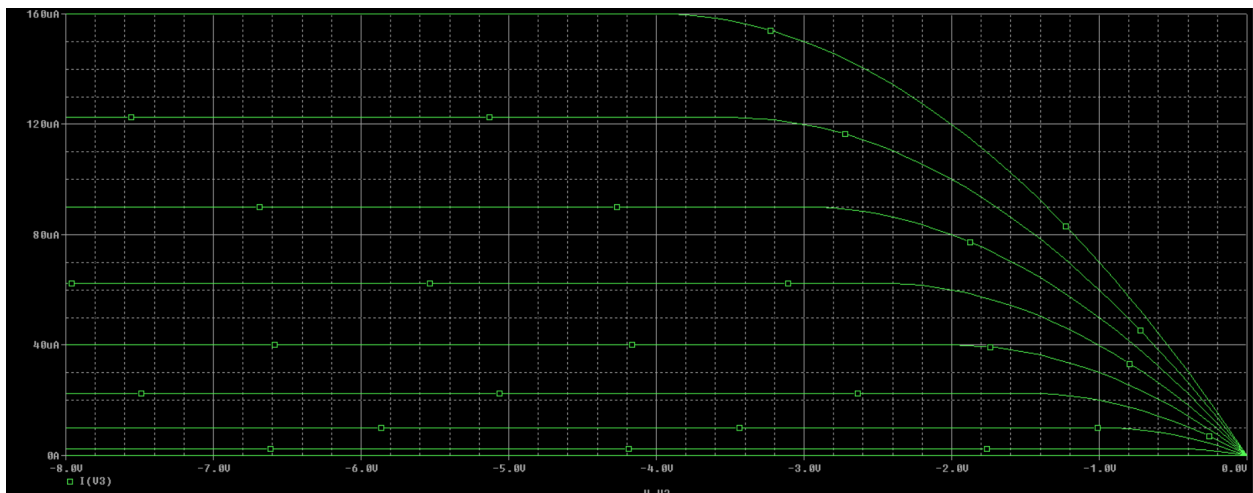
PMOS:

Goal of this experiment is to obtain the PMOS transistor output characteristics, i.e., drain to source current (I_{DS}) as a function of drain to source voltage (V_{DS}) for various gate to source voltages (V_{GS}) .

Sweep V_{DS} voltage in the range from 0 V to -8 V at an increment of -0.1 V.

Sweep V_{GS} voltage in the range from 0 V to -4 V at an increment of -0.5 V.

Record the characteristics.



Conclusion: This experiment was also a success due the NMOS and PMOS graphs matching the graphs that Professor Naher provided.

EXPERIMENT #3: Determine NMOS voltage transfer characteristics.

NMOS:

NMOS transistor Voltage Transfer Characteristics

Goal of this experiment is to obtain the NMOS transistor voltage transfer characteristics, i.e., the output voltage (V_{DS}) as a function of the input voltage (V_{GS})

Connect the NMOS transistor as shown in the Figure 3 below.

Set V_{DD} voltage at 8 V.

Set R_D equal to $1000\ \Omega$. Measure R_D _____

Set the V_{GG} voltage as per the values shown in the table.

At each V_{GG} value measure the drain to source voltage.

Record the values in this table.

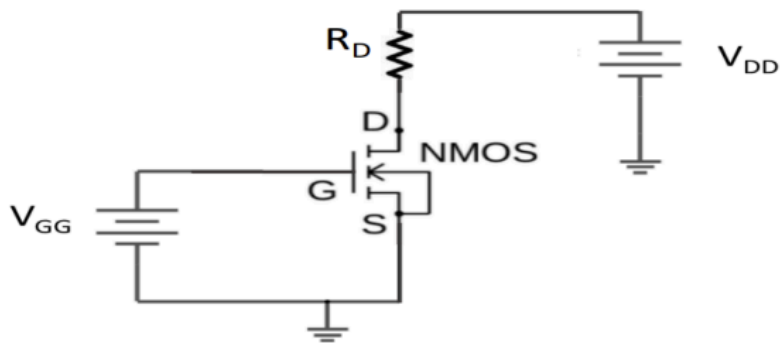
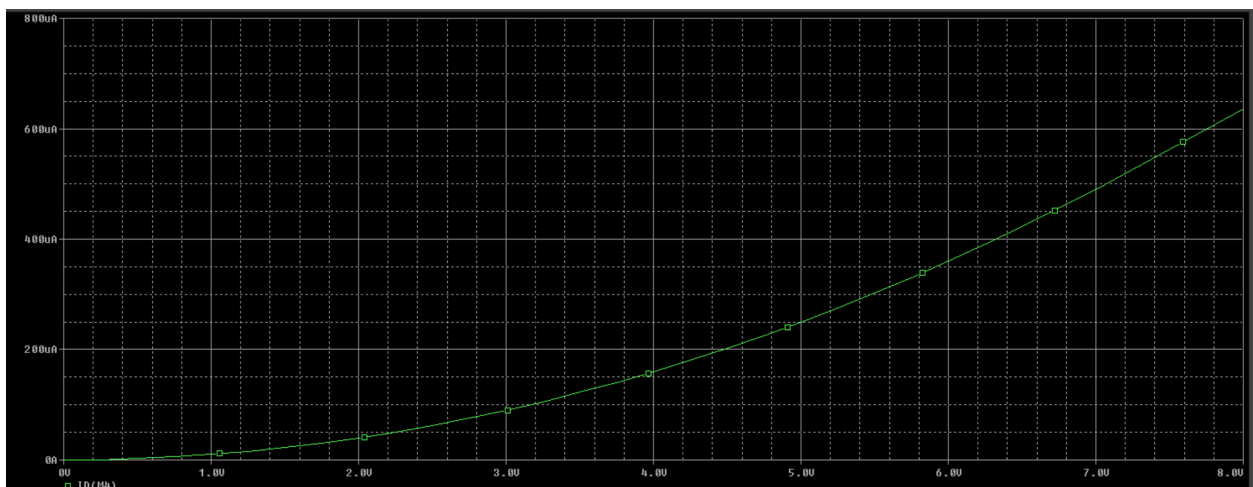
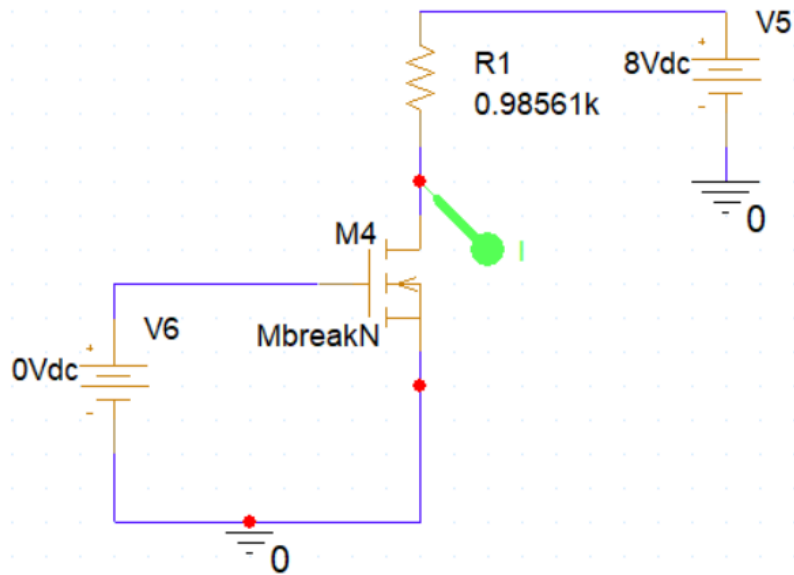


Figure 3

Experiment 3			
NMOS=2n7000FS			
R=0.98561KOhm			
	VGG (V)	Actual VGG	IDS
	0	0V	0A
	0.25	0.26912V	0A
	0.5	0.51919V	0A
	1	1.00552V	0A
	1.25	1.2633V	0A
	1.5	1.5257V	0A
	1.75	1.7581V	0.016mA
	2	2.0309V	0.4025mA
	2.25	2.2857V	4.7152mA
	2.5	2.5444V	8.1044mA
	2.75	2.7972V	8.1488mA
	3	3.0961V	8.1644mA
	3.25	3.2957V	8.1700mA
	3.5	3.5414V	8.1734mA
	4	4.0519V	8.1772mA
	4.25	4.3069V	8.1782mA
	4.5	4.5490V	8.1784mA
	4.75	4.8101V	8.1789mA
	5	5.0860V	8.1795mA
	6	6.0807V	8.1816mA
	7	7.0994V	8.1876mA
	8	8.0343V	8.1890mA





Conclusion: This experiment is also a success due to the exponential graph matching the values in the excel graph.

EXPERIMENT #4: Design & characterize NMOS Common Source Amplifiers

(a) Draw the load line Determine Operating Point Q and load resistor R_D

The goal of this experiment is to obtain the NMOS transistor output characteristics using a curve tracer. These characteristics show the drain-to-source current (I_{DS}) as a function of drain-to-source voltage (V_{DS}) for various gate-to-source voltages (V_{GS}). These measurements are similar to the previous experiment but the increment in the V_{GS} voltage sweep is smaller.

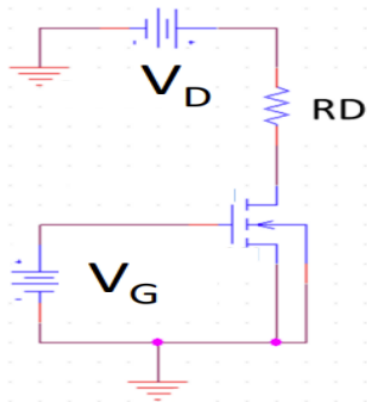
Use $R_D=500\text{ ohm}$.

Sweep V_{DS} voltage in the range from 0 V to +9 V, at an increment of 0.1 V.

Sweep V_{GS} voltage in the range from +1.5 V to +3.5 V at an increment of +0.25 V.

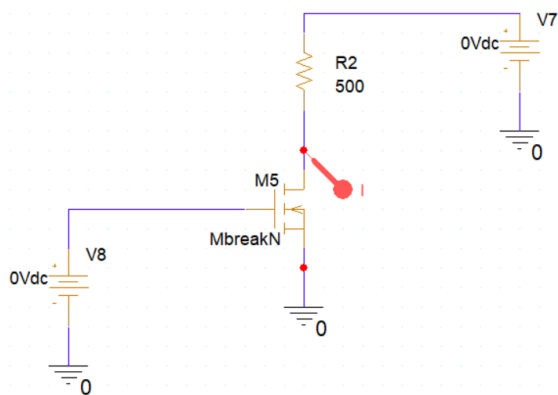
Record the characteristics PSpice.

What is the amplification PSpice?



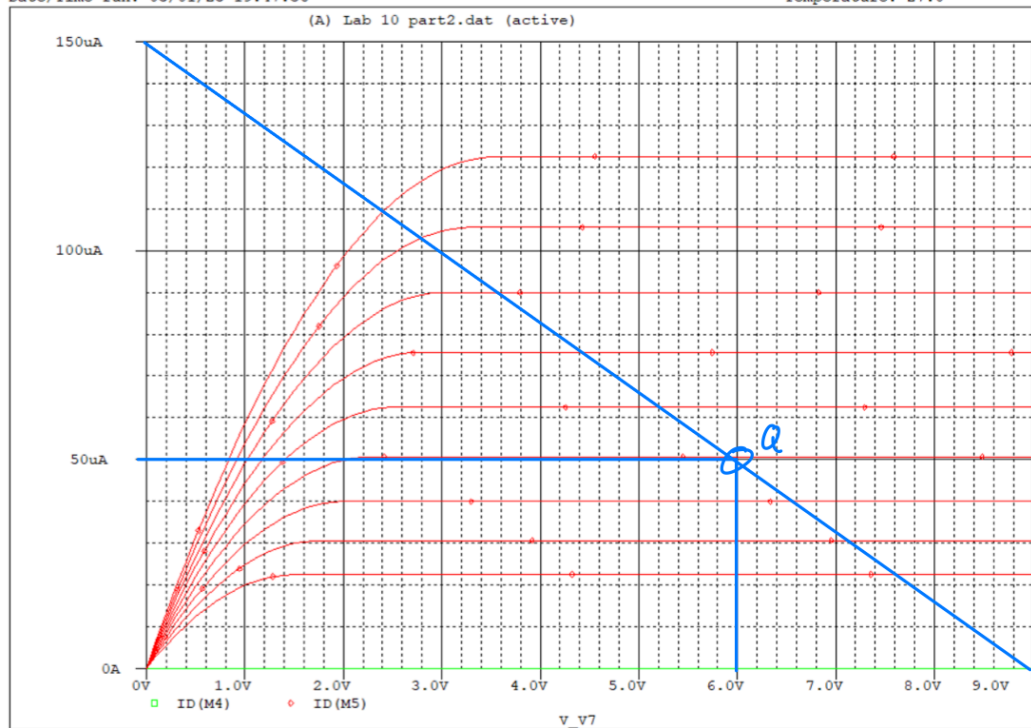
Now modify the circuit, $R_D = 20\text{k ohm}$ and $V_{DD}=20\text{v}$

V12



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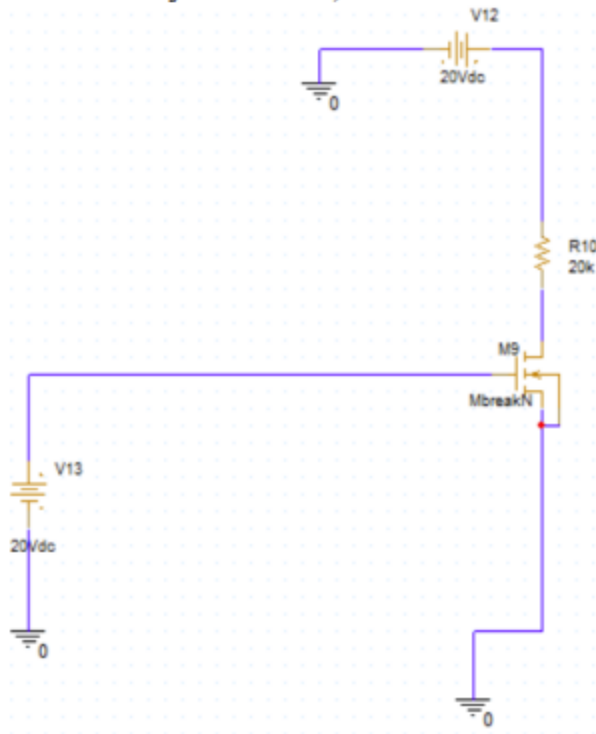


Date: May 01, 2025

Page 1

Time: 19:47:52

Now modify the circuit, $R_D = 20\text{k ohm}$ and $V_{DD} = 20\text{v}$

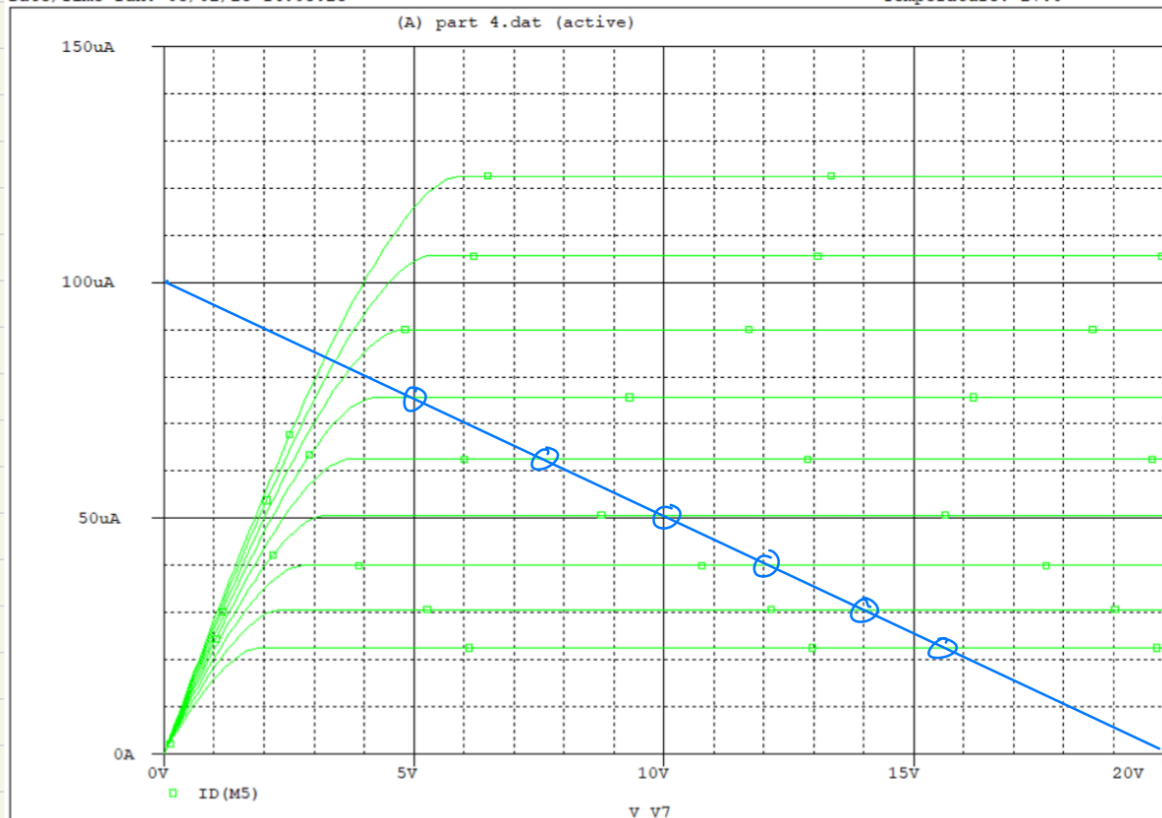


Record the characteristics using PSpice.

What is the amplification using PSpice?
Determine $V_{GSQ} = ?$

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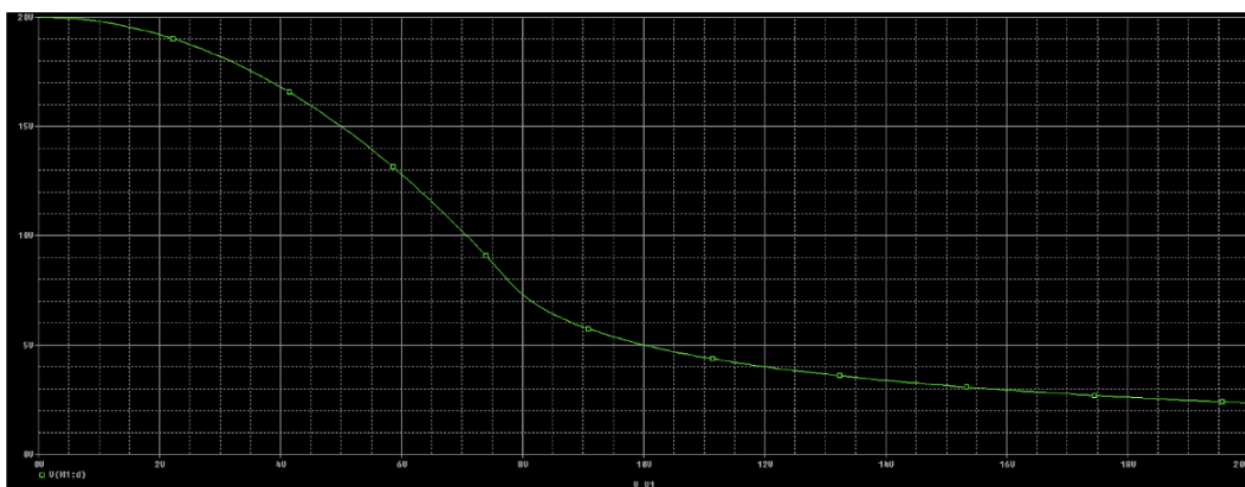
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Date: May 02, 2025

Page 1

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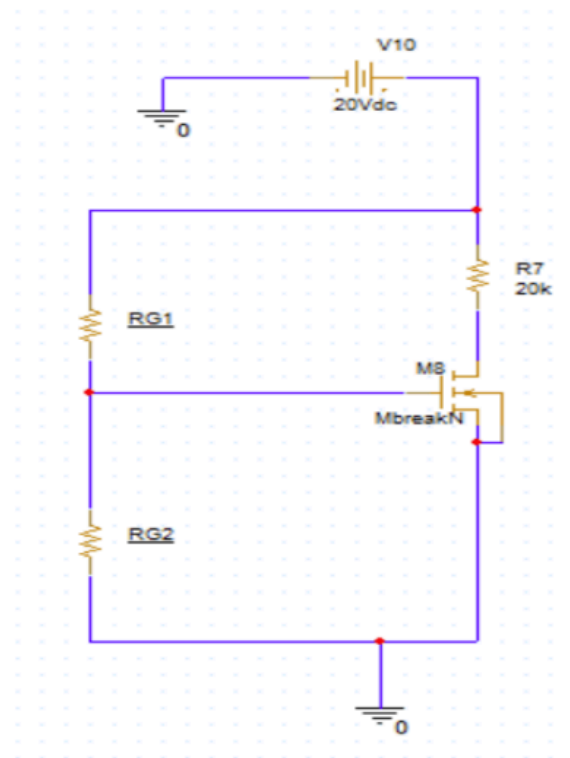


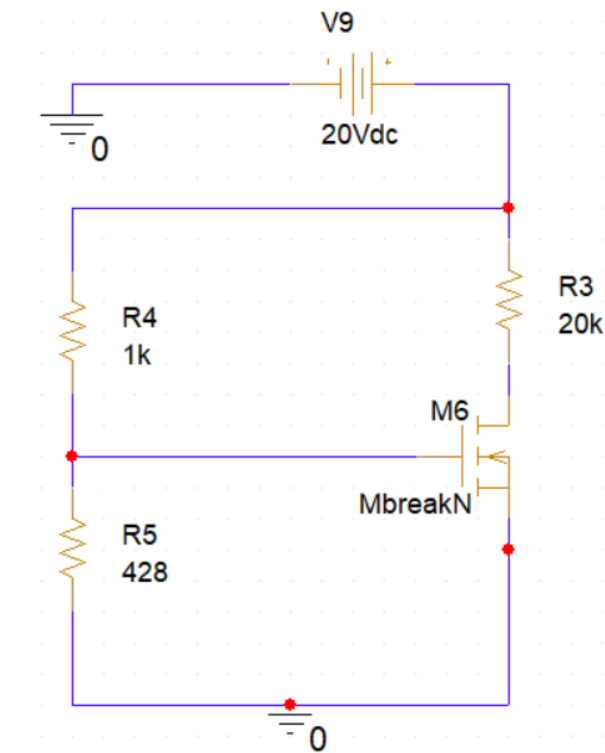
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		3.8453	0.000	3.8453						
	CURSOR 1,2	V(M1:d)	17.042	20.000	-2.9578	0.000	0.000	20.000	17.042	18.521

(b) Determine Bias Resistors RG1 and RG2 for each load line

You will build and characterize the amplifier circuit shown below.

Determine the values for resistors RG1 and RG2 that would set a VG bias at the gate approximately equal to the one that would correspond to its operating point Q, as determined in Experiment 4(a).

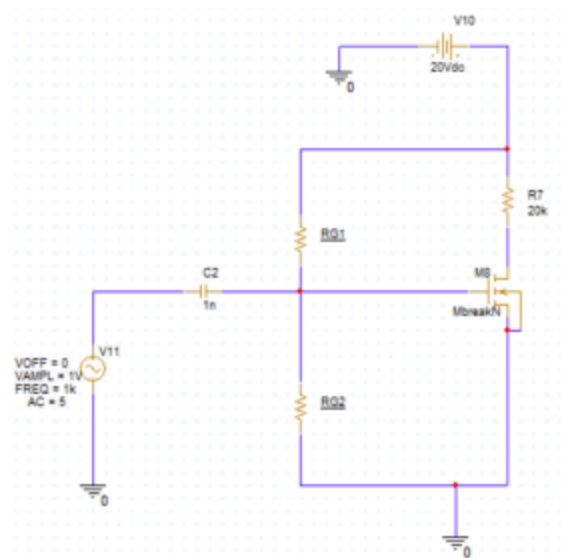




R1 Calculated	1Kohm
R2 Calculated	428ohm
R1 actual	0.98662Kohm
R2 actual	0.38734Kohm

$$V_G = 5.54 \text{ V}$$

(c) Build and Characterize Each of the Amplifier Circuits



Place a coupling capacitor C1 having a high value, e.g. 1nF, as shown. below.

Connect a sinusoidal signal source as shown and set the frequency of it to 10 kHz

Using a multimeter measure and record values of: $V_{GS} = \underline{\hspace{2cm}}$ $V_{DS} = \underline{\hspace{2cm}}$

$R_{G1} = \underline{\hspace{2cm}}$ $R_{G2} = \underline{\hspace{2cm}}$ $R_D = \underline{\hspace{2cm}}$

Record the input and output waveforms.

Calculate the gain.

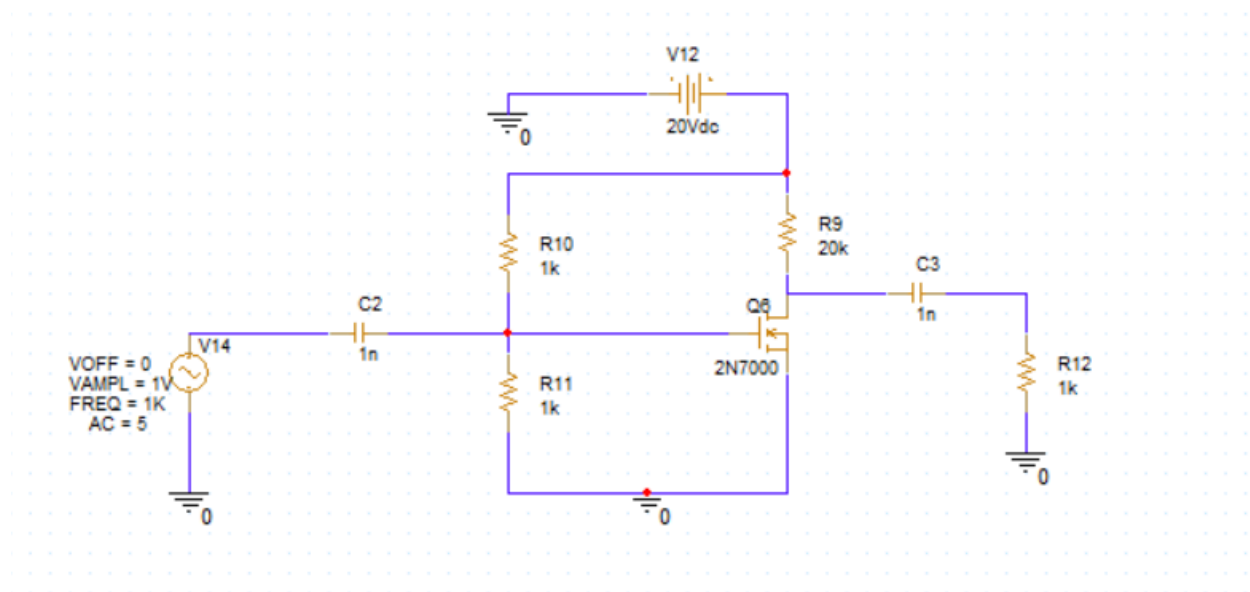
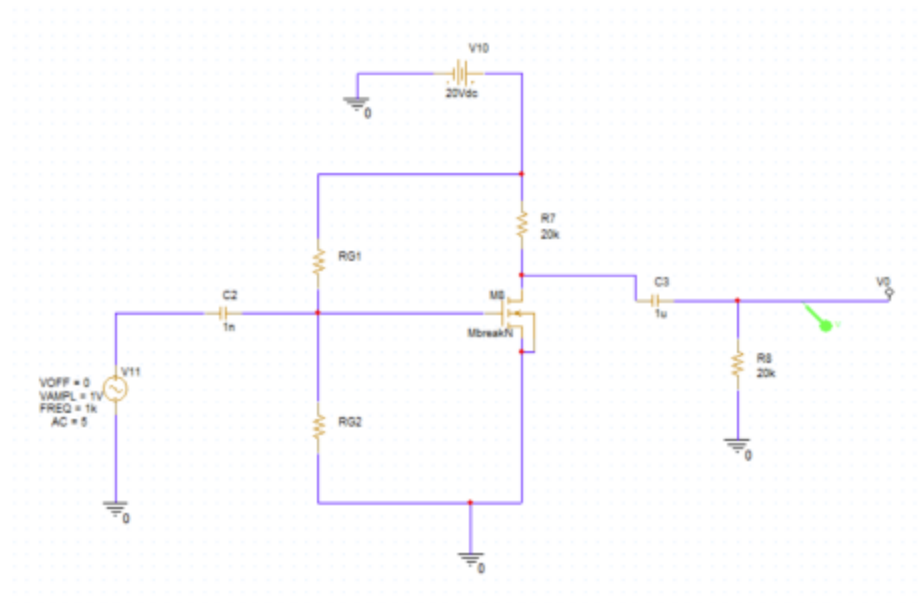
R1 Calculated	1Kohm
R2 Calculated	428ohm
R1 actual	0.98662Kohm
R2 actual	0.38734Kohm
R7	21.192Kohm
Vgs=5.729V	
Vds=3.919V	

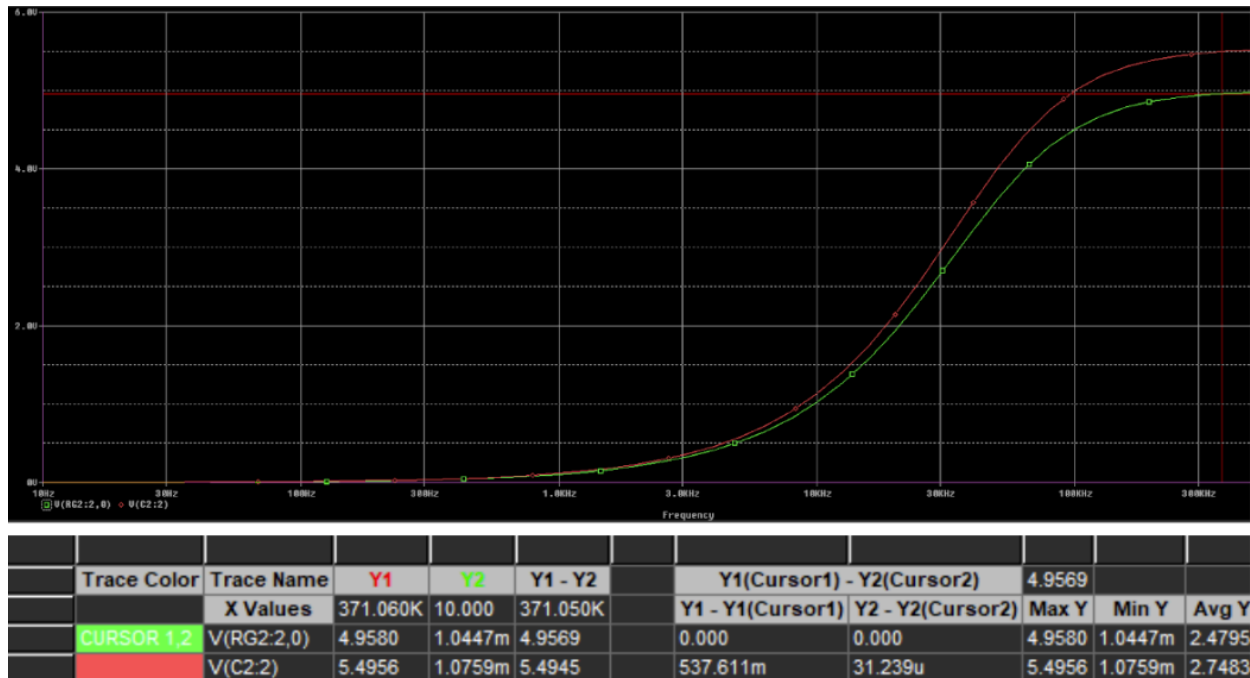
Input: Sine wave, 10kHz, 100 mV_{pp}

Output: Inverted sine wave, 10kHz, 4V_{pp}

Gain = -40dB

d) Add a high pass filter and do an AC sweep analysis (PSpice).





Conclusion: This experiment was also a success due to the close correlation between the measured and simulated results.